BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 251 · CPU components and registers

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which data structure supports last-in-first-out Undo?

Q2. What is the extra space complexity of allocating a new length-n array?

Q3. What does the ALU perform?

Q4. Simplify (A ∧ ¬B) ∨ (A ∧ B) and explain why the grouped cells make one variable disappear.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

ALU — Arithmetic Logic Unit.

CU — Control Unit, coordinates and decodes instruction operations.

MAR — Memory Address Register.

MDR — Memory Data Register.

CIR — Current Instruction Register.

PC — Program Counter.

ACC — Accumulator, stores intermediate arithmetic results.

New learning

Understand the CPU areas — The control unit decodes and coordinates operations; the ALU performs arithmetic and logical operations.

Separate address from data — MAR holds the address being accessed; MDR holds the instruction or data transferred; PC holds the next instruction's address.

Trace the simplified fetch — With PC=120, MAR becomes 120; memory supplies instruction via MDR to CIR; PC advances to the next sequential address. Branching may change PC later.

Worked example

If PC starts at address 50, trace register roles for the next sequential instruction fetch.

1. Predict first — Copy PC=50 into MAR to identify the memory address being accessed.

2. Explain the answer — Fetch memory content into MDR, copy it to CIR and advance PC to 51.

We Do · Practise with guidance

If PC initially holds 200 and Memory[200] contains ADD 10, state MAR, CIR and next sequential PC.

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MAR=200; CIR=ADD 10; PC=201
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Predict first

MAR must contain 200 and MDR transports the instruction from memory.

Explain the answer

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

CIR now contains ADD 10; for sequential execution PC has advanced to 201.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State the roles of PC, MAR, MDR, CIR and ACC.

Question 2

Trace a fetch starting PC=73 where Memory[73] contains LDA 92. State each register's role and the sequential next PC.

Question 3

Explain why the CPU uses both MAR and MDR rather than a single undifferentiated register, referring to an instruction fetch and a data load.

Exit Ticket · Answer before checking

The MAR stores the contents of the memory location being fetched.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A stack.

Q2. O(n).

Q3. Arithmetic and logical operations.

Q4. Factor A: A∧(¬B∨B)=A∧1=A. K-map adjacent cells differ only in B, so B is eliminated.

You Do pages · Independent questions

Q1. [5] PC next instruction address [1], MAR accessed address [1], MDR transferred data/instruction [1], CIR current instruction [1], ACC intermediate arithmetic/logical result [1].

Q2. [6] PC initially 73 [1], MAR becomes 73 [1], memory transfers LDA 92 via MDR [1], CIR receives LDA 92 [1], sequential PC becomes 74 [1], CU interprets the instruction [1].

Q3. [6] MAR carries memory location address [1]; MDR carries actual contents [1]; fetching uses PC→MAR [1] and memory→MDR→CIR [1]; loading data uses an address in MAR [1] and the retrieved value via MDR towards target register/ACC [1].

Exit Ticket · Model answer

False. MAR holds the address; MDR carries the contents.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding